



Purpose

In this experiment, you will utilize spectroscopic and gravimetric methods.

We will address the following questions:

1. What is the chemical formula of a hydrate with an unknown ratio of water?
2. Which technique is more useful for determining the ratio of water in an unknown hydrate?

Laboratory Activities

- A. Planning Gravimetric Experiment
- B. Carrying out Gravimetric Experiment
- C. Planning Spectroscopic Experiment
- D. Carrying out Spectroscopic Experiment
- E. Group Analysis and Argumentation

Laboratory Skills

- Weigh solids.
- Dilute solutions.
- Pipette liquids.
- Record absorbance spectra.

Introduction

Gravimetric Analysis and Conservation of Matter

Gravimetric analysis simply means *measuring the mass of a substance to determine its quantity*. It is an analytical technique that can take a variety of forms, such as quantifying the amount of a precipitate like you did in the Precision, Accuracy, and Precipitation Reactions experiment. Because of molar mass and Avogadro's number, we are easily able to convert between the number of particles, moles of substances, and mass of substances. This means that in the lab, we can measure masses using a balance and directly calculate the number of moles of that substance present. The Law of Conservation of Matter is what dictates the relationship between mass and moles present in a sample before and after physical or chemical changes. We will use this concept to explain the results of both experimental methods.

Ionic Solids: Hydrates vs. Anhydrous

Many ionic solids are hygroscopic, meaning they easily take up water into their crystal structure. When purchasing chemicals, there are often options to purchase an anhydrous solid or the hydrated form. The ionic formula portion is the same between the two compounds, but the presence or absence of water differentiates them.

At times, it is useful to have the anhydrous form, such as for water sensitive chemical reactions or to use as a desiccant (a substance that removes water). When a solid is going to be used in an aqueous solution, it does not really matter whether the anhydrous or hydrate form is used, but it will affect the molar mass of the compound and therefore the quantity that must be weighed out to make a specific molarity solution.

The hydrate form of an ionic solid can be heated to remove water, essentially converting it into the anhydrous compound. This technique could be an easy solution to save money or time waiting for a chemical delivery if all you have on hand is the hydrate version but need the anhydrous compound. Questions to consider: Why is this method possible? What happens to the water molecules upon heating? What are the typical melting/boiling points of ionic solids?

Spectroscopic Analysis

Spectroscopy, as we have already learned, refers to techniques that use light to detect chemical species. Visible absorption spectroscopy utilizes visible light to detect the absorbance of chemical species at specific wavelengths. Because visible absorption spectroscopy follows the Beer-Lambert law, it is very simple to convert absorbance values into concentration values, just like you observed in the previous lab.

Generating a calibration curve from known concentration standard solutions is the first step toward determining an unknown concentration of the same substance. Recall that the wavelength of maximum absorbance (λ_{max}) must be held constant for the value of the molar absorptivity (ϵ_{max}) to remain constant. The equation relating absorbance to concentration (Beer-Lambert law) is shown below with the error term included.

$$A = \epsilon cl + \delta \quad (\text{Equation CC.1})$$

Compounds Under Investigation

The compounds we will investigate in this experiment are copper(II) sulfate with an unknown hydrate ratio ($\text{CuSO}_4 \cdot \text{XH}_2\text{O}$), iron(II) sulfate with an unknown hydrate ratio ($\text{FeSO}_4 \cdot \text{XH}_2\text{O}$), and cobalt(II) nitrate with an unknown hydrate ratio ($\text{Co}(\text{NO}_3)_2 \cdot \text{XH}_2\text{O}$).

Calibration curves for each of the ionic compounds are provided below. Since water is not a species that can absorb visible light (it is clear), the concentrations shown refer to the ionic portion of the compounds only.

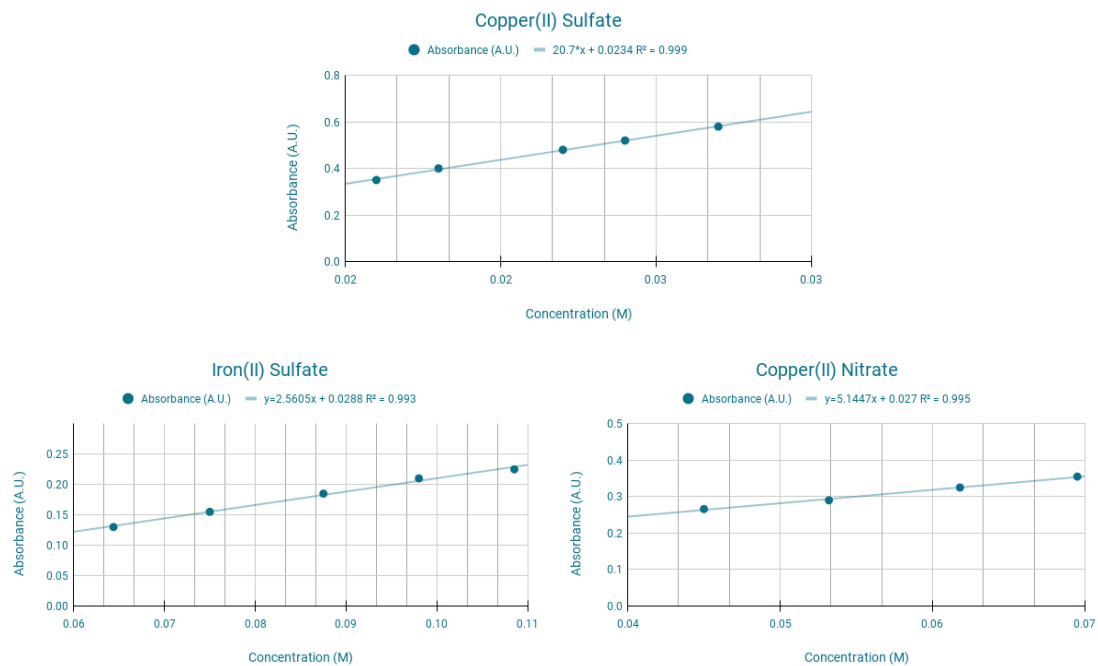


Figure CC.1: Calibration curves for each of the three ionic compounds under investigation..

Procedure

Reagents Available

- $\text{CuSO}_4 \cdot \text{XH}_2\text{O}$
- $\text{FeSO}_4 \cdot \text{XH}_2\text{O}$
- $\text{Co}(\text{NO}_3)_2 \cdot \text{XH}_2\text{O}$

A. Planning Gravimetric Experiment

1. As a group, select a compound to work with and record its formula.
2. With your group, determine the purpose of the gravimetric portion of the experiment. Reread the relevant background information and discuss together.
3. Determine a plan to measure the mass of the hydrate and anhydrous species and to relate mass to moles. Keep in mind that the heating process could take a while. Use approximately 1 gram total sample. Important things to consider while planning:
 - a. Moles are related to mass when the chemical formula is known
 - b. Any difference in mass between the hydrate and anhydrous compound can be attributed to water
 - c. How can you know when the sample is fully anhydrous?
 - d. Which form do you expect to weigh more?
 - e. What are some safety considerations?
4. Plan the data that will be collected and construct an appropriate table.

B. Carrying out Gravimetric Experiment

1. Follow your plan to measure the mass of both forms of your compound.
2. Be careful to use crucibles and tongs for any heating and/or picking up hot objects.
3. While waiting for the sample to change from hydrate form to anhydrous form, predict the mass of water that would be present in a 1 g sample of your product assuming a mole ratio of
 - a. 1:5 ionic solid:hydrate
 - b. 1:6 ionic solid:hydrate
 - c. 1:7 ionic solid:hydrate

C. Planning Spectroscopic Experiment

1. With your group, determine the purpose of the spectroscopy portion of the experiment. Reflect on how spectroscopy was used in the previous lab.
2. Determine a plan to prepare and measure the absorbance of a solution of a 15 mg/mL solution and to relate mass to moles.
 - a. Recall that absorbance is related to moles using Beer-Lambert relationships
 - b. Moles are related to mass
 - c. Any discrepancy in the measured and calculated masses can be attributed to the hydrates in the formula
 - d. Plan the data that will be collected and construct an appropriate table.

D. Carrying out Spectroscopic Experiment

1. Follow your plan to measure absorbance of the solution. Don't forget to calibrate the spectrometer with water before adding your sample to the cuvette holder!
2. Save the .csv file for your absorbance data and share with the group.

E. Group Analysis and Argumentation

1. Determine the mass of water in each sample. Discuss together and show your work with the stoichiometric calculations clearly written out.
2. Convert this mass into a chemical formula that replaces the X with a coefficient to represent the mole ratio of water in your chemical compound.
3. Develop an argument about both research questions with appropriate evidence and reasoning.
4. Share your argument with another group who investigated the same compound and discuss similarities and differences in your conclusions. Refine your argument as needed based on the conversations with the peer group.

Table CC.1: Group Argument

Claim 1	Claim 2
Evidence	Evidence
Reasoning	Reasoning